Gene 980 (2026) 149957

Contents lists available at ScienceDirect

Gene

journal homepage: www.elsevier.com/locate/gene

Review

Genetic and epigenetic determinants of injury risk and recovery in elite athletes: toward precision sports medicine

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ARTICLE INFO

Keywords:
Sports Medicine
Genomics
MicroRNAs
Oxidative Stress
Inflammation

ABSTRACT

Musculoskeletal injuries and interindividual variability in recovery remain critical challenges for sustaining elite athletic performance. Despite advances in diagnostics, training, and rehabilitation, unexplained differences in injury risk and recovery efficiency persist. Growing evidence implicates genetic and epigenetic mechanisms in these disparities, influencing collagen integrity, inflammatory regulation, oxidative stress, and muscle regeneration. Key contributors include single-nucleotide polymorphisms (SNPs) in COL1A1, COL5A1, ACTN3, and IL6, as well as novel loci identified by genome-wide association studies (GWAS). Polygenic risk scores (PRS), which integrate the effects of multiple low-impact variants, enhance prediction compared to single-gene markers. Epigenetic regulators, including DNA methylation and exercise-responsive microRNAs (e.g., miR-206, miR-133a, miR-486), provide a dynamic interface linking genetic predisposition with environmental factors such as training load and nutrition. Integration of genomic data with wearable technologies and artificial intelligence enables real-time monitoring, adaptive recovery, and precision load management. Yet, clinical translation faces challenges including limited population diversity, interpretative complexity, and ethical concerns over data privacy and potential misuse. Advancing precision sports medicine will require multidisciplinary, systems-level strategies that unite molecular insights with practical implementation, ultimately enabling individualized injury prevention, optimized rehabilitation, and the preservation of long-term athlete health and performance.

1. Introduction

In elite sports, injury prevention and optimized recovery are fundamental to sustaining high performance and extending athletic careers. Musculoskeletal injuries, particularly ligament tears, tendinopathies, and muscle strains, remain major causes of time loss, performance decline, and early retirement (Saragiotto et al., 2014; Kiel et al., 2025; Gimigliano et al., 2021; Bagherzadeh et al., 2025). Despite advances in training technologies and rehabilitation protocols, substantial interindividual variability in injury susceptibility and recovery efficiency persists, suggesting the influence of intrinsic biological factors beyond biomechanical load and environmental exposure. The term "elite athlete" in this review refers to people who have participated in national or international competitions and have completed at least five years of organized, high-performance training. Any musculoskeletal or soft-tissue impairment that results in changed performance or time lost longer than twenty-four hours is considered an injury. The pace and thoroughness of physiological recovery after exercise-induced stress or injury is referred to as "recovery efficiency." "Non-responders" are athletes who, despite standardized training, show minimal improvement in physiological adaptation or performance. Genetic variation has emerged as a key determinant in this context. SNPs in collagen, muscle, and cytokine-related genes (e.g., COL1A1, ACTN3, IL6) have been linked to tissue integrity, inflammation, and repair dynamics (Stone et al., 2020). While individual variants confer modest effects, PRS aggregate thousands of SNPs to improve the prediction of injury susceptibility and recovery potential (Wray et al., 2021; Pickering and Kiely, 2019). However, their clinical translation remains constrained by

Abbreviations: SNPs, Single-Nucleotide Polymorphisms; COL1A1, Collagen Type I Alpha 1; COL5A1, Collagen Type V Alpha 1; ACTN3, Alpha-Actinin-3; IL6, Interleukin 6; GWAS, Genome-Wide Association Studies; PRS, Polygenic Risk Scores; miR, MicroRNA; DNA, Deoxyribonucleic Acid; RNA, Ribonucleic Acid; MMP3, Matrix Metalloproteinase-3; CRISPR-Cas9, Clustered Regularly Interspaced Short Palindromic Repeats-Cas9.

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https://doi.org/10.1016/j.gene.2025.149957
Received 2 October 2025; Received in revised form 7 December 2025; Accepted 11 December 2025
Available online 12 December 2025
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limited predictive power, ethnic bias, and contextual variability. Beyond static genetic predispositions, epigenetic mechanisms provide a dynamic regulatory layer linking gene expression to training, nutrition, and circadian rhythms. DNA methylation and exercise-responsive microRNAs (e.g., miR-206, miR-486) influence inflammation, muscle repair, and adaptation to mechanical stress, offering promising but underexplored biomarkers for individualized recovery (Voisin et al., 2015). Together with transcriptomic, proteomic, and metabolomic insights, multi-omics approaches are beginning to reveal systems-level pathways underpinning adaptation and resilience in athletes (Hoffman, 2017). Recent work has additionally removed the boundaries between the applications of multi-omics in the monitoring of elite athletes in precision sports medicine. Various next-generation models, combining the power of genomic, pharmacogenomic, wearable biosensing, and multi-omics profiling, were proposed for the optimal personalization of individual athlete performance, recovery, as well as injury prevention strategies in elite sports (Juginović et al., 2025). The multi-omics data, derived from actual training conditions, has exhibited functional variation in the levels of particular metabolites, such as those of 4-hydroxyproline, methionine, oxaloacetate, as well as tyrosine, as well as the differential expression of particular proteins associated with the immune response, muscle injury, metabolic fitness, as well as hemostasis, offering molecular signatures of adaptation and recovery (Malsagova et al., 2024). Complementarily, the emerging "sportomics" paradigm integrates genomics, metabolomics, and proteomics with AI-driven analytics to characterize whole-body metabolic responses under real-time competitive conditions, enabling n-of-1 athlete profiling and personalized intervention strategies (Muniz-Santos et al., 2023). These advances underscore that multi-omics now operates beyond basic discovery science and has become a foundational component of precision athlete management.

Despite these advances, key gaps remain, including inconsistent definitions of "non-responders," methodological heterogeneity, and insufficient validation across diverse populations. Moreover, ethical and legal concerns regarding data privacy, consent, and potential misuse complicate clinical implementation (Webborn et al., 2015). The majority of studies that are currently available are based on relatively small cohorts, are largely dominated by Caucasian or sport-specific populations, and have limited generalizability despite the growing interest in genetic and epigenetic determinants of injury susceptibility. These limitations limit the transferability of polygenic or machine learning-based predictive models across various athletic groups and decrease the stability of effect-size estimates. As a result, current genetic associations should be viewed as probabilistic rather than deterministic, and further multi-ethnic validation is still required before they can be translated. This review synthesizes current evidence on genetic and epigenetic determinants of injury risk and recovery in elite athletes, evaluates the utility of polygenic models and gene, environment interactions, and highlights translational opportunities and challenges in precision sports medicine.

2. Scope and literature Search

The literature for this review was retrieved from PubMed, Scopus, and Web of Science using combinations of keywords including genetics, sports injuries, recovery, epigenetics, and nutrigenomics. Articles published in English between 2000 and 2025 were considered, with emphasis on studies involving elite athlete cohorts. Both candidate gene approaches and genome-wide association studies were reviewed. Due to the heterogeneity of the available evidence, the article is presented as a narrative synthesis rather than a systematic review.

3. Genetics of injury susceptibility

Injury susceptibility in elite athletes is a multifactorial trait influenced not only by extrinsic factors such as training load and biomechanics, but also by intrinsic biological determinants, particularly genetic variations. Understanding the genetic architecture underlying musculoskeletal vulnerability enables stratified injury prevention and rehabilitation strategies. Key genetic contributors include structural genes affecting connective tissues, genes modulating inflammatory and regenerative responses to mechanical stress, and novel variants uncovered through GWAS (Kahya and Taheri, 2024; Laguette et al., 2021).

3.1. Connective tissue genes

Genetic polymorphisms in collagen-related genes have been widely investigated due to their role in tendon and ligament integrity (Rahim et al., 2016).

COL1A1 (rs1800012): COL1A1 encodes type I collagen, the principal fibrillar collagen providing tensile strength and structural integrity in tendons and ligaments. The COL1A1 rs1800012 polymorphism, located within the Sp1 binding site, has been one of the most extensively studied variants in ligament and tendon injury genetics. Contrary to earlier small-scale studies suggesting increased risk, a *meta*-analysis of six eligible studies involving 933 injury cases and 1381 controls demonstrated that the TT genotype exerts a strong protective effect against sports-related tendon and ligament injuries. Significant associations were reported both in the homozygous model (TT vs GG: OR = 0.17, 95 % CI: 0.08–0.35) and the recessive model (TT vs GT/GG: OR = 0.21, 95 % CI: 0.10–0.44), with the strongest effects observed for anterior cruciate ligament (ACL) injuries. These pooled data indicate that rs1800012 may enhance collagen fibril organization and confer reduced susceptibility to mechanical failure in high-load tissues, highlighting its relevance for injury-risk profiling in elite athletes (Perini et al., 2022).

COL5A1 (rs12722): The rs12722 polymorphism within COL5A1, an essential regulator of collagen fibril, has been extensively studied due to its relevance to soft-tissue structural integrity (Gkritzali et al., 2024). A *meta*-analysis of nine studies (1,140 cases; 1,410 controls) demonstrated a significant association between the TT genotype and increased susceptibility to tendon and ligament injuries using the recessive model (TT vs. TC + CC: OR 1.58; 95 % CI 1.33–1.89; $P < 0.00001$). Stratified analyses further confirmed elevated risk across Achilles tendinopathy, anterior cruciate ligament injury, and lateral epicondylitis, with the strongest effects observed in Caucasian populations (OR 1.59; 95 % CI 1.33–1.90) (Raleigh et al., 2009; Lv et al., 2017).

Although COL5A1 is primarily implicated in connective-tissue pathology, emerging athletic cohort data suggest broader implications for muscle-injury susceptibility. In professional football players, individuals with the TT genotype exhibited higher muscle injury incidence, implying that impaired collagen fibril organization may compromise musculotendinous resilience under high-intensity, multidirectional loading. COL5A1 rs12722 is associated with increased odds of tendon and ligament injury in several populations, particularly in Caucasian cohorts (Jacob et al., 2022).

MMP3 (rs679620): The MMP3 locus has also been investigated in relation to musculoskeletal soft-tissue injury. A two-cohort case control study genotyping rs679620, rs591058, rs650108, and the functional promoter variant rs3025058 reported significant associations for variants within this region. In the South African cohort (n = 234 ACL ruptures; n = 232 controls), rs3025058, known to be in high linkage with rs679620, was associated with increased susceptibility to Achilles tendinopathy (OR = 2.88; 95 % CI: 1.4–6.1; P = 0.012). In the Australian cohort (79 chronic Achilles tendinopathy cases; 195 controls), a 6A-G-C-G haplotype constructed from these variants showed a protective effect (29 % in controls vs. 20 % in tendinopathy cases; P = 0.037). These findings indicate that variation within the MMP3 gene region contributes to Achilles tendon injury risk, although it remains unclear whether the genotyped variants are themselves causal or whether they tag risk-conferring variants elsewhere within the MMP gene cluster on chromosome 11 (Munhoz et al., 2010; Gibbon et al., 2017).

3.2. Muscle Injury-Related SNPs

Genetic variants contribute to inter-individual variability in injury susceptibility, but their predictive value remains probabilistic rather than deterministic (Appel et al., 2021). Among the most studied candidates are variants in ACTN3, CKM, IL6, and AMPD1, which together highlight the interplay between structural resilience and inflammatory recovery dynamics in muscle injury.

ACTN3 (rs1815739): The gene ACTN3 encodes the sarcomeric protein α-actinin-3, which is primarily expressed in type II muscle fibers. Muscle contractions that are strong and explosive are produced by type-II muscle fibers. The R577X polymorphism leads to a premature stop codon and the absence of functional alpha-actinin-3 in XX homozygotes. While this variant is associated with enhanced endurance in some athletes, it also correlates with reduced muscle stiffness and resilience to high-intensity eccentric loading. XX individuals are shown to have higher serum creatine kinase levels post-exercise, indicative of greater muscle fiber disruption (Baumert, 2018; Kittilsen et al., 2021). Supporting these findings, a controlled eccentric-exercise study (N = 157) demonstrated that ACTN3 XX homozygotes exhibited significantly lower baseline CK levels compared with RX/RR genotypes ($p = 0.035$), indicating genotype-dependent differences in muscle metabolic status even before exertional loading (Clarkson et al., 2005). The role of ACTN3 as one of the crucial genetic factors influencing both susceptibility to injury and recovery abilities is well-supported by longitudinal data. In this regard, the five-year longitudinal study conducted on elite cross-country skiers found a significantly greater prevalence of ACTN3 RR genotype in highly proficient athletes than those who were non-athletic, thereby implying the gene's role in both muscular power and endurance-related capacities. Nonetheless, although genotype-related discrepancies in the dynamics of VO2peak did not attain significance, athletes with ACTN3 variants showed characteristic trends in their aerobic recovery adaptation profile (Mägi et al., 2016).

CKM (rs8111989): This SNP in the muscle-specific creatine kinase gene modulates energy buffering during muscle contraction. Quantitative evidence supports the role of CKM rs8111989 in muscle performance phenotypes. A *meta*-analysis of nine studies comparing athletic cohorts with controls demonstrated that power athletes exhibit a significantly higher frequency of the G allele (OR = 1.14, 95 % CI: 1.02–1.28, P = 0.03) and the GG genotype (OR = 1.54, 95 % CI: 1.24–1.91, P < 0.0001). In contrast, no significant association was observed in endurance athletes (G allele: OR = 0.95, 95 % CI: 0.85–1.06, P = 0.34; GG genotype: OR = 1.00, 95 % CI: 0.78–1.27, P = 1.00). These findings suggest that rs8111989 may influence high-intensity, power-dominant muscle function more than endurance capacity, supporting its potential role in modulating susceptibility to muscle damage under repeated, high-load contraction (Chen et al., 2017). The AA genotype has been linked with a higher risk of muscle damage due to reduced efficiency in ATP regeneration and sarcomeric stability under repeated contraction cycles (Ginevičienė et al., 2021).

AMPD1 (rs17602729): AMP deaminase regulates purine nucleotide cycling during high-intensity effort. The T allele reduces enzymatic activity, impairing recovery capacity in anaerobic sports. Rare TT homozygotes exhibit severely diminished enzyme activity, altered muscle energy flux, and reduced power output (Hafen et al., 2022). Recent quantitative data further supports the significance of the AMPD1 c.34C > T (rs17602729) polymorphism regarding exercise performance. A systemic review and *meta*-analysis of 20 different studies conducted across 11 countries with the participation of a further 5717 individuals revealed that the CC genotype was significantly over-represented among endurance athletes (Odds Ratio [OR] = 1.72; 95 % CI = 1.40–2.12; $p < 0.00001$) and power athletes (Odds Ratio = 2.17; 95 % CI = 1.69–2.78; p = <0.00001) relative to the control group of individuals not practicing sport. On the other hand, the CT genotype (Odds Ratio = 0.61; 95 % CI = 0.49–0.75; $p < 0.00001$) and the TT genotype (Odds Ratio = 0.43; 95 % CI = 0.19–0.97; $p = 0.04$) showed higher under-representation among endurance athletes and the respective genotypes of the power athletes displayed the same pattern of under-representations ($p = 0.00001$; $p = 0.007$), respectively. Notably, genotype distributions did not differ between endurance and power disciplines (OR range: 0.76–1.39; p = 0.47–0.72), indicating that the functional advantage conferred by the CC genotype spans both aerobic and anaerobic metabolic pathways. These findings suggest that rs17602729 may influence susceptibility to exercise-induced fatigue and impaired recovery, particularly in athletes relying on repeated high-intensity efforts (Kartibou et al., 2025). These variants may influence both injury risk and performance in power-based disciplines (Kalinin et al., 2021), though epidemiological replication is still limited.

Toward Polygenic Models: While individual SNPs such as ACTN3, IL6, or AMPD1 show modest and often inconsistent effects, combined analyses offer stronger predictive potential. A Total Genotype Score (TGS) combining ACE, ACTN3, COL5A1, and MCT1 variants significantly separated injured from non-injured athletes in a longitudinal pilot study of professional Italian football players (n = 64). The risk of suffering a muscle injury was 3.5 times higher for players whose TGS was below the threshold (cut-off = 56.2 a.u.) (OR = 3.5, 95 % CI: 1.8–6.8, $p < 0.001$) (Maestro et al., 2022). Similar findings in academy football players, where multiple SNPs including ACTN3, ACE, VEGFA, and ACAN were linked to performance and injury phenotypes, further support the polygenic nature of muscle injury susceptibility. These candidate gene studies provide the foundation for genome-wide approaches aimed at uncovering additional loci with small but cumulative contributions to injury risk (McAuley et al., 2021).

A summary of key candidate genes associated with injury susceptibility, including both connective tissue–related and muscle injury related SNPs, is presented in Table 1.

3.3. GWAS findings in musculoskeletal injuries

GWAS offer a powerful, hypothesis-free approach to identify SNPs linked to complex traits, including susceptibility to musculoskeletal injuries (Hettiarachchi and Komar, 2022). A recent athlete-specific GWAS in a high-performance Estonian National Team cohort (n = 126; 82.5 % male) provided detailed population and phenotype-stratified insights into performance and injury genetics. Athletes were categorized by performance (medalists n = 29 vs. non-medalists n = 97) and by injury status (injured n = 47 vs. non-injured n = 79), with additional stratification by injury subtype (patellar tendinopathy n = 22, Achilles tendinopathy n = 17, hamstring injury n = 3, ACL rupture n = 6). Three SNPs reached genome-wide significance for elite performance, including DSG1 rs10502567 (OR = 14.3) and DSG4 rs73410248 (OR = 17.4), while 76 loci showed suggestive significance. For musculoskeletal injuries, 37 SNPs demonstrated suggestive associations, particularly PAPPA2 rs11580456 (OR = 13.8) and MAS1 rs220735/rs170219 (OR ≈ 3.1). Additional injury-specific signals were observed for COL22A1 (rs3924862), PLXNA2 (rs11799530), DOK5 (rs73142922), GNG12 (rs28435277), and DAP (rs267959, rs2930047, rs1080440, rs267939). Collectively, these findings highlight that athlete-specific GWAS can identify high-effect, phenotype-specific variants that are not detectable in general-population studies and strengthen the genomic architecture underlying sports-related musculoskeletal injuries. By providing effect sizes, injury-specific stratification, and sport-specific phenotypes, the inclusion of athlete-specific GWAS, such as the Estonian National Team cohort, significantly improves the interpretability of genetic associations, addressing prior limitations of population-level GWAS for sports-injury prediction (Ebert et al., 2023). Unlike candidate gene studies, GWAS enables the discovery of novel genetic loci that contribute to an athlete's polygenic architecture of musculoskeletal resilience by examining large cohorts (Ebert et al., 2023). A recent *meta*-analysis of six studies reported no overall association between the VEGFA rs699947 polymorphism and tendon or ligament injury risk; however, in a European subgroup analysis after removing one heterogeneous study the

Table 1
Candidate Genes Associated with Injury Susceptibility in Elite Athletes.

Gene / Variant	Population (ancestry)	N (cases/controls)	Effect (model)	95 % CI	p-value	Reference
COL1A1 (rs1800012)	Mixed (*meta*-analysis of 6 studies)	933 cases / 1,381 controls	OR = 0.17 (TT vs GG; homozygote)	0.08–0.35	P < 0.001	(Chen et al., 2017)
COL5A1 (rs12722)	Predominantly Caucasian (*meta*-analysis; effect significant only in Caucasian subgroup)	1,140 cases / 1,410 controls	OR = 1.58 (TT vs TC + CC; recessive)	1.33, 1.89	P < 0.00001	(Lv et al., 2017)
MMP3 (rs679620)	South African cohort (case-control)	ACL cohort: 234 cases / 232 controls (ACL study) −Australian cohort: 79 cases / 195 controls (Achilles)	R = 2.88 (rs3025058, South African −Achilles)	1.4–6.1	P = 0.012	(Gibbon et al., 2017)
ACTN3 (R577X)	Estonian longitudinal ski cohort	58 skiers (5-year follow-up) + 322 controls	_	_	RR: p = 0.003;	(Mägi et al., 2016)
CKM (rs8111989 (A/G) power athlete status −allelic (G vs A))	Mixed populations (9 studies)	1,396 / 10,092	1.14	1.02–1.28	p = 0.03	(Chen et al., 2017)
CKM (rs8111989(A/G) Power athlete status genotypic (GG vs AA + AG))	Mixed populations	698 / 5,046	1.54	1.24–1.91	P < 0.0001	(Chen et al., 2017)
CKM(rs8111989 (A/G) Endurance athletes − allelic (G vs A))	Mixed populations	2,280 / 9,068	0.95	0.85–1.06	P = 0.34	(Chen et al., 2017)
CKM(rs8111989 (A/G) Endurance athletes − genotypic (GG vs AA + AG))	Mixed populations	1,140 / 4,534	1.00	0.78–1.27	P = 1.00	(Chen et al., 2017)
IL6 (−174G > C, rs1800795)	Healthy adults	70 healthy physical education students (42 males and 28 females; aged 22–32)	3.29	1.27–7.85	P = 0.009	(Yamin et al., 2008)
AMPD1 (rs17602729 (c.34C > T)(Endurance athlete status)	Mixed populations (meta, 11 countries)	5717 total (Exact cases/controls vary per study)	CC vs controls OR = 1.72 (endurance)	1.40–2.12	P < 0.00001	(Kartibou et al., 2025)
AMPD1 (rs17602729 (c.34C > T)(Power athlete status)	Mixed populations (meta, 11 countries)	5717 total (Exact cases/controls vary per study)	CC vs controls OR = 2.17 (power)	1.69–2.78	P < 0.00001	(Kartibou et al., 2025)
AMPD1 (rs17602729 (c.34C > T)(CT genotype)	Mixed populations (meta, 11 countries)	5717 total (Exact cases/controls vary per study)	CT reduces odds (endurance OR = 0.61; power OR = 0.51)	0.49–0.75 / 0.39–0.65	P < 0.00001	(Kartibou et al., 2025)
AMPD1 (rs17602729 (c.34C > T)(TT genotype)	Mixed populations (meta, 11 countries)	5717 total (Exact cases/controls vary per study)	TT strongly underrepresented (endurance OR = 0.43; power OR = 0.25)	0.19–0.97 / 0.09–0.68	P = 0.04 / 0.007	(Kartibou et al., 2025)

dominant model indicated a modest protective effect of the AA/AC genotypes compared with CC (OR = 0.92, 95 % CI: 0.86–0.98; P = 0.015). These findings suggest a potential genotype-dependent effect in European populations, although the sensitivity of the results to study selection and underlying heterogeneity warrants cautious interpretation (Varley et al., 2018; Li et al., 2024). Similarly, the functional SNP rs143383 in the 5′-UTR of GDF5 a growth differentiation factor essential for joint development has been robustly linked to early-onset osteoarthritis, supporting its potential role in joint and tendon pathophysiology among athletes (Southam et al., 2007).

However, replicating these findings across different cohorts remains essential. Heterogeneity in injury definitions and athlete types can confound associations. For example, several studies have reported inconsistent findings regarding collagen-related SNPs such as COL1A1 rs1800012 in relation to soft tissue injuries like hamstring strains. While a *meta*-analysis found the TT genotype to be protective, it noted variability across ethnicities and injury types (Guo et al., 2024), and a separate cohort of Italian soccer players failed to find a significant association with ACL rupture (Massidda et al., 2023). These discordant results highlight the need for sport-specific genetic profiling and robust phenotyping. Ultimately, the application of the data from GWAS in multivariable models, such as PRS, is needed for the advancement of prediction and personalization in injury prevention (Kujala et al., 2020).

3.4. Polygenic risk scores and predictive tools

Given the complex nature of injury susceptibility, single-gene based methodologies exhibit restricted predictive potential. In contrast, the PRS has emerged as an integrated tool by compiling the effects of multiple SNPs into a summative measure of risk (Jayasinghe et al., 2024). Initial research stipulates that integration of certain genotypes, such as ACTN3 R577X, COL5A1 rs12722, and IL6 −174G > C, may augment the predictive ability concerning susceptibility to musculoskeletal injuries; however, integrated models need investigation (Guo et al., 2022; Zouhal et al., 2023; Baumert, 2018). A chief drawback is the very complicated relationships brought about by genetic variations. For instance, an observed considerable correlation with greater injury risk was apparently found in one study with a Caucasian cohort that studied the heterozygous GT genotype of COL1A1 rs1800012, whereas candidate genes such as COL5A1 (rs12722 and rs13946) seem not to bear any level of significance; it is postulated that not all candidate genes may carry an equally significant relevance or that similar gene-gene interactions are feeble or contextual (Leźnicka et al., 2021). In consideration of the discussed multifaceted complexities, the integration of PRS with machine learning algorithms holds potential in refining predictive modalities towards greater precision and personal injury-risk prediction (Mamani, 2020). Translational relevance of the predictive modalities is evidenced by research outputs in other clinical fields, whereby inflammation-related PRS have effectively predicted conditions such as stroke with such impact being moderated with lifestyle (Park, 2024).

PRS models for athlete profiling, although promising, present several obstacles for athletes: poor transferability across populations; lack of standardized reporting; unwarranted levels of predictive accuracy in small cohorts (Wand et al., 2021). These limitations should be

appropriately dealt with through robust validation and individualized interpretation before proper implementation of PRS in sports-specifically for athlete applications.

3.5. Epigenetic contributions

Besides static DNA sequences, epigenetic changes DNA methylation, histone acetylation, and non-coding RNAs determinificantly impact gene expression in response to environmental stimuli such as exercise. By their very nature, epigenetic changes are dynamic and reversible. Thus, skeletal muscle undergoes rapid transcriptional reprogramming according to the demands posed by the trainings performed, injury, as well as recovery. While some of the reactions, like histone acetylation or microRNAs in response to exercise stimuli occur in the minutes-to-hours timeframe and involve acute responses, others like DNA methylation might occur for weeks to months. Such longer-term 'muscle memory' aspects are now being recognized by researchers to play pivotal roles in the dynamics of athletic adaptability (Nahumi et al., 2022; Bure et al., 2022; Ghanimi et al., xxxx).

An important concept is epigenetic muscle memory, in which certain DNA methylation modifications in skeletal muscle are retained after detraining and reactivated on retraining. This idea is supported by data from human genome-wide methylation analysis. Key hypertrophy-related genes (such as AXIN1, GRIK2, CAMK4, and TRAF1) are hypomethylated even after muscle mass returns to baseline during unloading, indicating preservation of previous training signatures, according to a seminal resistance-training study by Seaborne et al. Genes like UBR5, PLA2G16, SETD3, and HEG1 showed even more hypomethylation and transcriptional activation during subsequent reloading, matching increased hypertrophic gains. Significant evidence for long-term epigenetic imprinting was provided by the hypomethylation of several loci (including GRIK2, TRAF1, STAG1, and BICC1) that persisted for up to 22 weeks following a single resistance-exercise session (Seaborne et al., 2018). This proposes a maintained transcriptional advantage that might account for individualised adaptation and recovery response patterns (Pilotto et al., 2024). Evidence from acute exercise studies also demonstrates that epigenetic remodeling can occur within minutes. For example, a well-controlled human trial showed that a single bout of endurance exercise induces rapid DNA hypomethylation at promoters of key metabolic genes PGC-1α, PDK4, and PPAR-δ paralleling their dose-dependent increase in mRNA expression. Similar promoter hypomethylation was observed in mouse soleus muscle 45 min after contractile stimulation, supporting the concept that acute contraction triggers fast, reversible methylation changes that facilitate metabolic gene activation(Barres et al., 2012). Although skeletal muscle is still the tissue of greatest study, recent research also illustrates epigenetic remodeling of the heart and neurons after exercise, imputing systemic adaptation with recovery, stress response, and autonomic audit in athletes (Zheng et al., 2025). Elucidation of epigenetic responses provides new intervention points for maximising recovery and injury resilience. An integrated strategy that combines epigenetic markers with established blood-based indicators (e.g., creatine kinase, C-reactive protein) might provide added sensitivity in tracking athlete recovery and guiding precision load management between training cycles (Lee et al., 2017). Note, however, that these epigenetic modifications do not act in an independent manner. DNA methylation, for example, could potentially regulate the accessibility of chromatin, and in turn, the levels of histone acetylation. Similarly, the modification of histones affects the environment in the vicinity of the transcribed genes and regulates the expression of microRNAs. Some microRNAs, on the other hand, regulate the epigenetic enzymes, like the DNMTs and HDACs, and an intricate network, in the form of bidirectional feed-forward loops, results (Seaborne et al., 2018; Kouzarides, 2007).

3.6. Sex and age specific genetic effects

The genetic variation-injury risk interaction is strongly determined by both sex and age. For instance, the association of COL1A1 rs1800012 T allele with ACL injury risk is much stronger in female athletes, probably as a result of the influence of hormones on collagen metabolism (Sivertsen et al., 2019). Recent research highlights sex steroid hormones, especially estrogen, as key molecules regulating connective tissue integrity and inflammation response, and supports an explanation of the observed increased susceptibility of female athletes to ligament injuries (Hunter and Senefeld, 2024). This statement is also reinforced by an extensive study reporting significantly increased ACL rupture risk in female COL5A1 rs12722 TT genotype carriers, but a reduced one in their male equivalents (Wang et al., 2017) Age also mediates the impact of genetic factors. The pro-inflammatory IL6 −174C variant, for example, may contribute to impaired healing and prolonged inflammation in older athletes, compounding age-related declines in tissue regenerative capacity (Baumert et al., 2016). Furthermore, epigenetic profiling using DNA methylation clocks suggests that elite athletes may display accelerated biological aging, yet this is often accompanied by beneficial anti-inflammatory and anti-tumor epigenetic signatures (Spólnicka et al., 2018).

These differential effects underscore the critical need for age- and sex-specific genetic profiling in sports medicine. Integrative frameworks should combine candidate gene findings, GWAS data, and polygenic models with demographic and temporal modifiers, such as sex-specific hormonal profiles and training periodization (Murtagh et al., 2023). These insights are foundational for developing effective personalized prevention, monitoring, and rehabilitation protocols.

4. Genetic influences on recovery and adaptation

There is considerable inter-individual variation in the recovery process following severe exercise-induced damage (acute recovery) and the long-term reaction that results in enhanced function or structural healing (chronic adaptation). Genetic polymorphisms are important factors that affect molecular pathways associated with both acute and chronic processes, such as long-term tissue remodeling, training adaptation, and overall injury resilience, as well as oxidative stress management, inflammation resolution, and rapid tissue repair kinetics. For elite athletes to maximize recovery techniques and ensure precise load management, understanding these genetic inputs is essential(Ghanimi et al., xxxx; Pickering and Kiely, 2019).

4.1. Inflammatory response genes

Those genes at the core of the inflammatory cascade control centrally the rate and the speed of restitution of muscle damage (Yamin et al., 2008).

IL6 (rs1800795): Inflammatory reactions are regulated and coordinated by the cytokine IL-6, which has both pro- and anti-inflammatory functions. Skeletal muscle cells are among the various cell types that make it.The −174G > C promoter polymorphism modulates IL-6 transcription and plays a dual role in inflammation-driven recovery and high-intensity performance. Carriers of the C allele exhibit elevated post-exercise IL-6 and creatine kinase (CK) concentrations following eccentric contractions, indicating prolonged inflammation and slower recovery kinetics. These genotype-dependent responses have been replicated in trained athletes, where the C allele is consistently associated with heightened muscle soreness and delayed restitution after load-bearing exercise (Yamin et al., 2008).

Conversely, in elite competitive cohorts ($n = 253$), the G allele shows functional relevance to neuromuscular power production: the GG genotype was significantly overrepresented in power athletes compared with endurance athletes and non-athletic controls ($P \approx 0.03$). Individuals with the GG genotype demonstrated a 2.47-fold greater

likelihood of being classified as elite power athletes (OR = 2.47; 95 % CI: 1.24–4.92). Collectively, these findings indicate that IL6 rs1800795 exerts context-specific effects where the C allele is linked to exaggerated inflammatory responses and delayed recovery, whereas the G allele favors short-duration explosive performance. These insights highlight the importance of integrating inflammatory-genetic and performance-related pathways when interpreting IL6-related susceptibility to muscle injury and training adaptation (Serpa et al., 2024).

TNFα (rs1800629): One powerful pro-inflammatory cytokine that contributes to the acute phase stress response is tumor necrosis factor-alpha (TNF-α). The G > A polymorphism at position −308 of the TNFα gene, located within the gene promoter region, controls transcription activity so that the A allele may elevate the expression of TNF. Increased TNF may have an adverse impact upon muscle adaptation via the dampening of muscle cell fusion and differentiation. Furthermore, a shift toward chronic effects is evident as the polymorphism was linked with decreased improvements in physical performance in older adults, and chronic elevation of TNF is thought to prevent long-term muscle remodeling (Baumert et al., 2016).

CRP (rs1205): The SNP rs1205 located within the CRP gene exhibits an association with CRP variations, manifestations of systemic inflammation. Carriers of the C allele are found with elevated baseline CRP levels (Dehghan et al., 2011). While there are no direct correlations with post-exercise acute responses, the variant's clear relationship with chronic systemic inflammation may hamper muscle remodeling processes, particularly for older individuals (de Santis et al., 2020). There is, however, some evidence that exercise may offset the deleterious influence of the allele and therefore suggest an interaction potential between genetic factors and lifestyle choices (Liu et al., 2017) These polymorphisms may be utilized as biomarkers to create individualized recovery plans and periodized training schedules (Rahim et al., 2019; Pickering and Kiely, 2019).

4.2. Oxidative stress response genes

Intensive physical exercise causes the accumulation of reactive oxygen species (ROS) with potential muscle fatigue and mitochondiral damage as an outcome (Wang et al., 2021). The effectiveness of antioxidant protection is determined genetically, and specifically SNP on major genes (Krishnamurthy et al., 2024).

SOD2 (rs4880): Val16Ala substitution (rs4880) in SOD2 gene influence scavenging activity of enzyme. Ala allele augments mitochondrial translocation and antioxidant activity, providing enhanced cellular protection (Pourvali et al., 2016). The impact of the variant seems sport-specific; there was no significant correlation with power athletes, yet it was attributed to performance in endurance athletes (Kazan et al., 2023), indicating that Val/Val carriers might be at an increased risk of oxidative injury and reduced recovery.

GPX1 (rs1050450): The Pro198Leu substitution (rs1050450) in the GPX1 gene reduces the enzymatic activity of glutathione peroxidase 1, an enzyme critical for detoxifying hydrogen peroxide (Eskafi Sabet et al., 2014; Jablonska et al., 2009). This can predispose carriers of the Leu (T) allele to higher oxidative stress during strenuous training, which may impair adaptive responses. Although some studies have not found a significant effect, this may be due to the underrepresentation of the disadvantageous Leu/Leu genotype in healthy athletic populations (Kotowska and Jówko, 2020). These genetic variations highlight the importance of considering gene-environment interactions, particularly in endurance sports where oxidative stress is a key factor in performance and recovery.

4.3. Circadian Rhythm genes and recovery patterns

Circadian genes govern daily physiological rhythms that influence hormone secretion, sleep, tissue repair, and overall acute recovery efficiency. Variations in these genes can affect the synchronization of internal recovery processes with external cues. Core clock genes, such as CLOCK, BMAL1, and their targets like CRY1/2, modulate key metabolic and anabolic pathways (e.g., AMPK, mTOR, PGC-1α) that regulate protein synthesis, glycogen resynthesis, and mitochondrial biogenesis. Disruption in these pathways can acutely delay muscle regeneration and, over time, impair training adaptation (Aoyama and Shibata, 2017; Gutierrez-Monreal et al., 2024; Viggars et al., 2024).

CLOCK (rs1801260): Specific polymorphisms in these genes are linked to recovery outcomes. For example, the T allele of CLOCK (rs1801260) is associated with a delayed sleep phase and reduced melatonin secretion, which can influence anabolic hormone release and muscle repair during sleep (Haraguchi et al., 2019; Bandín et al., 2013; Xu et al., 2010).

PER3 (VNTR polymorphism): The PER3 gene's VNTR polymorphism also affects sleep structure; individuals with the 5-repeat allele show enhanced slow-wave sleep and improved physical recovery, while 4-repeat carriers may experience greater fatigue after heavy training loads (Viola et al., 2007).

CRY1/CRY2: CRY1 and CRY2, as core circadian repressors, regulate metabolic pathways and synchronize peripheral clocks in skeletal muscle. Muscle-specific Cry1/2 knockout models demonstrate disrupted energy homeostasis, diminished aerobic capacity, and reduced motor performance, underscoring their pivotal role in coordinating tissue repair and energy availability in response to training stimuli (Mansingh and Handschin, 2022; Unconfirmed 376721, xxxx).

Beyond structural and inflammatory pathways, circadian regulators play an essential role in recovery efficiency and adaptation. Clinical evidence shows that daily oscillations of core-clock gene expression in blood, saliva, and hair correlate with muscle properties and exercise capacity, with strength and endurance performance typically peaking in the late afternoon (15:00–18:00) (Basti et al., 2021). These rhythms influence not only metabolic and physiological processes but also psychological domains such as alertness and mood, which are shaped by chronotype, hormone fluctuations (cortisol, testosterone, melatonin), and prior light exposure (Pradhan et al., 2024).

Environmental factors also interact with circadian regulation. For example, in a Chinese Han population, the PER2 rs2304672 polymorphism moderated the effect of job stress on sleep disturbances, with G allele carriers showing higher susceptibility (Wang et al., 2022). This gene-environment interaction illustrates how genetic susceptibility and lifestyle factors converge to shape recovery and performance. In practical terms, non-invasive sampling of circadian biomarkers (e.g., saliva, blood) could guide training schedules and recovery protocols tailored to an athlete's chronotype. Looking forward, integrating circadian loci into polygenic recovery scores may allow more precise and individualized optimization of recovery timelines (Nelson et al., 2025; Augsburger et al., 2025).

4.4. Genetic contribution to chronic adaptation and structural integrity

Recent longitudinal and genomic evidence further substantiates the genetic contribution to recovery and adaptation processes. A five-year follow-up study in elite cross-country skiers investigated ACE I/D and ACTN3 R577X polymorphisms in relation to aerobic performance dynamics. The frequencies of the ACE ID and ACTN3 RR genotypes were significantly higher among male skiers compared with controls, suggesting a potential advantage for combined power and endurance traits. Although genotype-related differences in VO2peak changes over time were not statistically significant, the observed trends indicate a possible genotype-dependent adaptation pattern to prolonged training stimuli.

Complementarily, a genome-wide association study in orthopedic patients identified 52 genetic variants associated with implant loosening after total joint arthroplasty. Notably, variants within IFIT2/3, CERK, and PAPPA genes showed the strongest associations, with odds ratios ranging from 12.6 to 21.6. Genotype-dependent variability in the time to aseptic loosening ranged from 8 to 16 years, underscoring a substantial

genetic influence on long-term tissue stability. Although derived from a clinical rather than athletic population, these findings conceptually reinforce the contribution of genetic variation to tissue repair, mechanical resilience, and recovery potential in human musculoskeletal systems.

5. Epigenetics and Injury/Recovery

Epigenetic modifications heritable yet reversible changes in gene expression without DNA sequence alteration are increasingly recognized as key regulators of injury risk and recovery. Acting as an interface between genetic predisposition and environmental stimuli such as exercise, sleep, stress, and nutrition, they provide a molecular basis for individual variability in adaptation. Among the best-studied mechanisms are DNA methylation and exercise-responsive microRNAs, the latter of which will be examined in detail in the following subsection (Tarnowski et al., 2022; Bittel and Chen, 2024). Long-term training induces stable methylation patterns in skeletal muscle, shaping transcriptional responses to exercise. For example, endurance-trained athletes show lower promoter methylation at MYH7 (slow-twitch fibers) and higher methylation at regulatory genes such as PGC-1α, consistent with an "epigenetic memory" that facilitates long-term adaptation (Geiger et al., 2024). These patterns of methylation can also give us a clue regarding age-related differences in methylation and their involvement in recuperation and disease susceptibility.

In addition to the epigenetic memory of training, elite sports would also appear to affect biological aging. Relative to non-athletes, power athletes exhibited a larger "epigenetic age" based on hypermethylation of CPG sites at TRIM59 and KLF14 genes, which have significant roles in anti-tumor and anti-inflammation activities. Of particular relevance, these epigenetic modifications may initiate self-protection signaling networks, thus explaining partially the reduced risk of cardiovascular diseases and some tumors among elite athletes (Spólnicka et al., 2018).

5.1. DNA Methylation: A molecular memory of exercise

DNA methylation, the attachment of methyl groups at CpG cytosines, regulates transcription in a context-specific fashion. Transcription is typically repressed by promoter hypermethylation, especially at CpG island promoters. Gene-body and enhancer methylation, however, have more nuanced, even activating, effects (Jones, 2012). Acute endurance exercise rapidly induces promoter hypomethylation at key metabolic regulators such as PPARGC1A (PGC-1α), PPARD (PPAR-δ), and PDK4, thereby facilitating transcriptional activation and metabolic remodeling (Barres et al., 2012). With repeated training, a subset of these changes consolidates into longer-lived signatures, consistent with a skeletal-muscle "epigenetic memory." Genome-wide profiling of vastus lateralis biopsies across training, detraining, and retraining phases has shown persistent hypomethylation at genes involved in myogenesis and extracellular matrix (ECM) remodeling, which coincides with enhanced hypertrophic responses during retraining (Seaborne et al., 2018).

Emerging evidence suggests that such methylation dynamics partly explain interindividual variability in exercise adaptation and recovery. While single-site effects are modest, cumulative methylation patterns can reflect training history, influence tissue resilience, and shape inflammation and remodeling processes. Understanding these dynamics provides a foundation for personalized exercise strategies and identifies potential molecular targets to optimize muscle function and long-term health (Voisin et al., 2015).

5.2. microRNAs in muscle Injury, Recovery, and adaptation

MicroRNAs (miRs) represent a fundamental post-transcriptional regulatory step whose coordination of processes such as inflammation, muscle regeneration, ECM repair, and metabolism following exercise-induced injury represents their physiologic role. Contrary to past models describing their role primarily in exercising distinct pathways, it now appears these biomolecules have their specificity influenced by their origin (intracellular versus circulating) and physiologic state of training versus injury. Most evidence on exercise-responsive microRNAs derives from association studies, and while mechanistic experiments suggest regulatory roles, their causal contribution to injury or adaptation in humans remains to be clarified (Nahumi et al., 2022; Giordano et al., 2020; Zhang and Chen, 2018).

5.2.1. Myostatin/TGF-β signaling (miR-23a, miR-27a)

Among the most prominent muscle remodeling-regulating miRs, the core regulators in the myostatin/TGF-β pathway are represented by miR-23a and miR-27a. Both miR-23a and miR-27a have been shown in experimental models to target and suppress MSTN, a key inhibitor of muscle growth. Such interactions are associated with hypertrophic and antifibrotic responses following exercise (Fernández-Sanjurjo et al., 2018). These miRs are extremely muscle-cell enriched. After resistance and eccentric exercises, their levels get stimulated (Wang et al., 2017).

5.2.2. Satellite cell activation and myogenesis (miR-206)

miR-206 is implicated in regulating satellite cell activation and myogenic differentiation, partly through its interactions with Pax7 and MyoD. MiR-206 is expressed in response to muscle damage in order to coordinate very effectively the kinetics of muscle regeneration. However, in athletic individuals who are often regularly subject to 'microtrauma,' the levels of the 'classic' muscle-related small RNA miR-206 are usually elevated (Zabihi and Akhoondian, 2025).

5.2.3. Metabolic and IGF-1/Akt signaling regulation (miR-486)

miR-486 is one of the most functionally defined muscle-enriched microRNAs and represents a crucial integrator of anabolic signaling for the recovery of metabolism post-exercise injury. miR-486 has been associated with enhanced IGF-1/Akt signaling, potentially through the repression of PTEN and FoxO1 in experimental systems. Exercise research indicates the consistent induction of miR-486 post-resistance exercise in association with the stimulation of muscle hypertrophy programs but reduced levels in scenarios of chronic fatigue, overreaching, or overtraining (Wang et al., 2017; Qiu et al., 2024). While intracellular miRNAs regulate muscle-specific processes such as myogenesis, inflammatory resolution, and mechanotransduction, only a subset of these molecules is released into circulation via exosomes or protein-bound complexes. Importantly, circulating levels do not always mirror intracellular expression, which is a major consideration when evaluating miRNAs as non-invasive biomarkers of muscle damage and recovery.

5.2.4. Intracellular versus circulating miRNAs

The fact that intracellular miRNA responses do not always reflect their circulating levels is a crucial distinction for the development of biomarkers. In muscle tissue, intracellular miRNAs mainly represent local signaling programs related to myogenesis, inflammation, and metabolic remodeling. On the other hand, circulating myomiRs like miR-1, miR-133a, and miR-206 tend to rise after systemic stress reactions and muscle fiber damage, though this relationship is not strictly proportional and depends on the type of exercise, the timing of the sampling, and the release dynamics specific to the fiber. Evidence currently available suggests that intracellular and circulating miRNA kinetics partially overlap, but this correspondence is context-dependent and continues to be a major obstacle to their translational use as biomarkers (Soplinska et al., 2020).

5.2.5. Performance and clinical significance

When taken as a whole, exercise-responsive miRNAs control basic biological processes that control muscle recuperation, vulnerability to injury, and adaptability. Their incorporation into multimodal biomarker panels may improve training load monitoring, help identify the risk of

overuse-related injuries early, and support customized recovery and periodization plans. Establishing standardized measurement frameworks and validating their clinical utility in athletic populations will require more work.

these epigenetic factors play coordinated roles. The specific mechanisms and functional outcomes are summarized in Table 2.

Whereas PRS reflect static genetic predispositions, exercise-responsive miRNAs serve as dynamic, environmentally modulated biomarkers. Their integration may provide a more precise framework for predicting injury risk and guiding recovery strategies (Fig. S1).

Circulating miRNAs (ci-miRNAs) have emerged as promising non-invasive biomarkers for monitoring athletes' adaptive responses to training and overall health status. Beyond their diagnostic potential, ci-miRNAs regulate central pathways such as the IGF1/PI3K/AKT/mTOR axis, thereby linking external training stimuli to molecular mechanisms of muscle anabolism and recovery. Their expression's manifestation varied according to training type and training intensity, thus acting as active indicators of physiological adaptation. Therefore, it is important to determine certain ci-miRNA signatures and their respective targets in order to encourage evidence-based, personalized approaches in sports medicine (Li et al., 2025).

5.3. Clinical implications and future Perspectives

Inclusion of epigenetic profiles in personal sports medicine holds new promise for exercise performance optimization and prevention of risk of injuries. Exercise-responsive miRNAs and methylation patterns may be monitored to correlate molecular outcomes with implementable athlete management techniques. For example, studies have suggested that training-related adaptations are tracked by responsive biomarkers in the form of circulating miRNAs (c-miRNAs) that correlate with endurance and strength athlete categories (Wardle et al., 2015). By increasing basic knowledge of underlying mechanisms, these molecular findings hold promise for development of personal training and recovery programs. Application of state-of-the-art multi-omics technologies (genomics, transcriptomics, proteomics, and metabolomics) enables measurement of molecular shifts at several levels of biology simultaneously and, thus, detailed training response understanding, muscle adaptation, and recovery processes. By use of this knowledge, specific interventions may be introduced toward exercise performance optimization, prevention of maladaptive response and overtraining, and optimization of return-to-play plans (Hoffman, 2017).

Nevertheless, application of omics technologies in the field of exercise research is faced with significant challenges related to data acquisition, computational processing, interpretation, and integration of molecular dimensions. These issues will require systems interdisciplinary methodologies that integrate physiology, molecular biology, and informatics. In spite of these deterrents, such systems hold promises for unraveling new molecular targets and advancement of precision exercise science. Moreover, future studies applying multi-omics approaches are poised to reveal dynamic regulatory networks governing resilience and regeneration in elite athletes (Hoffman, 2017).

Table 2
Role of Key Exercise-Responsive MicroRNAs (Ex-miRs) in Skeletal Muscle Repair and Adaptation.

Clinical and Scientific Utility	Biological Outcome in Muscle Repair	Primary Regulatory Function (Mechanism of Action)
miR-206	Upregulated in response to injury; Promotes the transition from myoblast proliferation to satellite cell differentiation (Myogenesis).	Crucial for the structural and functional recovery of the damaged Neuromuscular Junction (NMJ).
miR-133a/b	Highly expressed in muscle; Regulates myoblast proliferation and inhibits programmed cell death (Apoptosis).	Essential for maintaining and expanding the pool of muscle precursor cells (MPCs) necessary for tissue repair.
miR-486	Targets negative regulators of the Insulin-like Growth Factor 1 (IGF-1)/Akt signaling pathway.	Significantly supports muscle anabolism, hypertrophy, and functional adaptation to exercise load.

5.4. Limitatins and Technical challenges

Understanding the molecular underpinnings of exercise-induced adaptation is advancing rapidly; however, substantial methodological and biological challenges still limit the translation of these findings into elite sports practice. These limitations span from experimental design constraints to analytical standardization issues, and they differ depending on the type of molecular marker under investigation. For clarity, key challenges can be grouped into three main domains: DNA methylation studies, circulating microRNA biomarkers, and multi-omics integration (Roberts and Markby, 2021).

5.4.1. DNA methylation studies

A review of 25 human studies by Voisin et al. demonstrated that exercise-induced DNA methylation is highly tissue- and gene-specific, limiting generalization across cohorts. Interpretation is further hampered by small sample sizes and heterogeneity in exercise protocols, intervention length, and sampling methods (Voisin et al., 2015). Even advanced platforms such as the Infinium MethylationEPIC BeadChip, which predominantly profile vastus lateralis muscle, fail to capture methylation responses in other relevant tissues. Moreover, multi-phase designs (training, detraining, retraining) add analytical complexity, complicating the identification of phase-specific effects (Seaborne et al., 2018). Together, these issues underscore the need for standardized methodologies and larger longitudinal cohorts.

5.4.2. Circulating microRNA biomarkers

Research on circulating miRNA biomarkers faces challenges of diagnostic specificity and reproducibility. Reported inconsistencies largely stem from pre-analytical variability in sample collection, processing, and storage, as well as the complexity of miRNA transport through carriers such as extracellular vesicles. The absence of standardized protocols for isolation, quantification, and normalization further limits cross-study comparability. To advance clinical utility, future work should prioritize harmonized analytical pipelines, targeted evaluation of miRNA carriers, and exploration of alternative biofluids beyond blood (Witwer, 2015).

5.4.3. Multi-Omics integration

Integrating multi-omics data (genomics, epigenomics, transcriptomics, proteomics, metabolomics) poses major computational and logistical challenges. These efforts require standardized, high-quality multi-tissue datasets and robust pipelines to manage large, heterogeneous data. Progress further depends on cross-disciplinary collaboration, data-sharing frameworks, and validated analytical workflows, as emphasized by Hoffman (Hoffman, 2017). Increasing evidence demonstrates that integrating multiple omics layers genomics, transcriptomics, proteomics, and metabolomics substantially enhances the predictive resolution of injury-risk models and individual recovery profiles. Recent frameworks such as enduromics and resistomics emphasize that endurance and resistance training elicit distinct molecular signatures measurable across multiple biological layers, providing individualized metabolic and inflammatory fingerprints that can refine training prescription and reduce injury risk. Likewise, multi-omics studies of resistance-exercise rehabilitation have shown that coordinated analysis of plasma proteomics, PBMC transcriptomics, and metabolomics can identify immune-modulatory pathways that mediate functional recovery, suggesting that multi-omics profiling captures mechanistic dimensions of adaptation that single-omic approaches miss. Together,

these findings highlight the translational value of multi-omics integration for developing more robust predictive models, improving athlete monitoring, and moving toward precision training and recovery strategies (Khoramipour et al., 2025; Zhou et al., 2025).

One of the main practical limitations regards cost/scalability considerations. Comprehensive multi-omics profiling particularly untargeted proteomics, metabolomics, and whole-genome or transcriptome assays remains considerably more expensive and operationally demanding than single-layer testing, restricting its use largely to research or high-resource elite settings. The extent to which the added predictive value justifies these costs requires evidence from validation studies demonstrating meaningful improvements in model performance and in clinically relevant outcomes such as reduced time-loss injuries. A pragmatic path forward includes: (i) rigorous evaluation of incremental predictive utility; (ii) development of lower-cost targeted panels derived from discovery multi-omics; and (iii) pilot cost-effectiveness analyses in representative athlete cohorts. Until such data are available, implementation should be cautious and focused on high-risk groups where potential benefits are more likely to outweigh testing costs (Wu et al., 2019; Chen et al., 2023).

6. Polygenic risk scores in sports medicine

PRS integrate the additive effects of numerous SNPs identified through GWAS, providing a more robust measure of genetic predisposition than single-gene markers. In sports medicine, PRS have shown potential in stratifying injury risk (e.g., tendon and ligament rupture, muscle strains) and predicting variability in training adaptation and recovery. Although PRS provide a useful framework for summarizing genome-wide genetic contributions to injury and performance phenotypes, their current utility for individual athlete stratification remains limited. First, the proportion of variance explained by existing PRS for exercise-related traits is modest, typically ranging between 0.07 % and 1.44 % in large cohorts such as the UK Biobank (Kujala et al., 2020). Second, most PRS have been developed primarily in European-ancestry datasets, resulting in reduced predictive accuracy and poor calibration when applied across diverse populations owing to differences in allele frequencies and linkage disequilibrium patterns (Duncan et al., 2019). Third, few PRS have undergone independent, sport-specific external validation or reported standardized predictive metrics such as area under the curve (AUC), variance explained (R^2), or odds ratio per standard deviation (OR per SD), limiting their clinical interpretability (Torkamani et al., 2018). Finally, important practical considerations including data privacy, informed consent, and the risk of genetic determinism indicate that PRS should be considered research tools rather than ready-to-use screening instruments in athlete management. Future progress will require large multi-ethnic sport-specific GWAS, harmonized injury phenotyping, and formal evaluation of predictive performance to establish clinical utility (Patel and Fahed, 2023; Choi et al., 2020).

6.1. Predictive Models

PRS have been applied in general populations to predict traits relevant to exercise physiology, such as osteoporosis, tendinopathy, and cardiovascular disease (Al Rifai et al., 2022; O'Sullivan et al., 2022). For example, Kujala et al. showed that PRS from UK Biobank data modestly predicted objectively measured physical activity levels (0.07–1.44 % variance), underscoring their potential translational value. In sports contexts, early-stage PRS have been proposed for injury risk stratification, including anterior cruciate ligament rupture, Achilles tendinopathy, and muscle strains, by aggregating variants across COL1A1, COL5A1, ACTN3, and MMP3 (Maffulli et al., 2013). Although the predictive power of individual SNPs is limited, composite PRS models demonstrate improved accuracy by integrating thousands of variants into a single score (Torkamani et al., 2018).

6.2. Integration into athlete health Management

Integrating PRS into athlete monitoring systems offers opportunities for personalized risk stratification and rehabilitation planning. Athletes with high genetic susceptibility could receive targeted preventive interventions such as neuromuscular training or tailored load management (Collins and Raleigh, 2009). The concept of a "recovery PRS" has also been proposed to refine return-to-play decisions and optimize rest schedules, moving rehabilitation toward individualized, precision-based protocols (Pickering and Kiely, 2019). Table 3 summarizes emerging applications of PRS in elite sport, illustrating their potential to move athlete management from generalized protocols toward genetically informed strategies.

Future advances in PRS utility will rely on larger, multi-ethnic cohorts, functional SNP annotation, and integration with transcriptomic and epigenomic data. While ethical concerns, privacy issues, and sport-specific validation remain critical challenges, the translational potential is substantial. Successful deployment could shift athlete care from reactive to predictive, reducing injury incidence and supporting long-term performance (Duncan et al., 2019; Hasin et al., 2017).

7. Gene-Environment interaction in sports recovery and injury prevention

Genetic tendencies influencing responses to a range of external stimuli, such as training volume, dietary intake, and recovery techniques, are revealed by gene environment (G × E) interactions. These are essential for precision in sport science and sport medicine, as well as for supporting athlete customization (Nieman, 2021; Spanakis et al., 2025). Such multi-layered interactions are schematically depicted in Fig. S2.

There are seasonal patterns of hormonal fluctuations which may, in turn, affect the efficiency of recovery for professional athletes. Experimental evidence obtained through animal models confirms high seasonal variability of physiological indicators, such as substantial growth hormone increases during the summer period (Koks et al., 2000). Although this research was conducted on animal models, evidence confirms the lack of constant growth hormone levels in all twelve months, following the same patterns as animal behavior and neuroendocrine processes. Considering the crucial contribution of growth hormones to tissue healing and collagen synthesis, seasonal shifts in hormonal patterns may affect genetic propensities toward injuries, adaptation, and recovery periods for professional athletes. However, limited population diversity and small sample sizes limit the predictive accuracy of polygenic scores and machine-learning frameworks, which may reduce model transferability across different ethnic groups and sporting disciplines.

7.1. Load Management + Genotype

Musculoskeletal adaptation, overtraining risk, and tolerance to training stress are all influenced by genetic variation. Gene variants affecting tissue resilience, fiber type composition, and inflammatory response, including COL1A1, ACTN3, and IL6, shape vulnerability to tendon or muscle injury under repetitive load. These results lend support to periodization strategies that are informed by genotype (Bulgay et al., 2023; Serpa et al., 2024) Crucially, nutrigenomic factors also influence optimal adaptation, emphasizing the necessity of incorporating genetics, load, and customized nutrition into recovery planning (Ahmetova-c et al., 2016).

7.2. Nutrigenomics in injury Recovery

The study of nutrigenomics looks at how genetic variations affect the metabolism and recovery processes of nutrients (Kiani et al., 2022). For example, gene variations in MTHFR (Raghubeer and Matsha, 2021)

Table 3
Applications of Polygenic Risk Scores (PRS) in Sports Medicine.

Study / Population	Polygenic Model Components (SNPs)	Phenotype Predicted	Sample Size (N)	Model Type	Performance Metrics	Reference
Chinese Elite Athletes (power vs endurance)	ACTN3 (rs1815739), ADRB3 (rs4994), PPARGC1A (rs8192678)	Power performance (identification & prediction)	Model construction: n = 103; External validation: n = 125	Logistic regression-based polygenic model	Predictive model: AUC = 0.736; Identification model: AUC = 0.854	(Yang et al., 2021)
World-class power vs endurance athletes	IL6 (rs1800795), NAT2 (rs1208), NOS3 (rs2070744)	Discrimination between elite endurance vs. power performance	Endurance = 100; Power = 53 (same ethnicity)	Multivariate logistic regression (polygenic discriminant model)	AUC = 0.72 (95 % CI: 0.66–0.81); Model χ^2 = 25.3, df = 3, p < 0.001; Variance explained = 21.4 %	(Buxens et al., 2011)
Spanish elite power vs. endurance athletes	ACE I/D, ACTN3 R577X, AGT Met235Thr, GDF-8 K153R, IL6 − 174G/C, NOS3 − 786 T > C	Polygenic prediction of elite power-oriented profile	Power = 53; Endurance = 100; Controls = 100 (all Spanish Caucasian males)	Total Genotype Score (TGS) model, max = 100	Mean TGS: Power = 70.8 ± 17.3 vs. Endurance = 60.4 ± 15.9 (P < 0.001); vs. Controls = 63.3 ± 13.2 (P = 0.012). Perfect TGS = 100: 9.4 % of power athletes vs. 0 % in other groups. Population probability: 0.2 % (~1 in 500 Spanish).	(Ruiz et al., 2010)

FTO, and PPARG have been established to affect folate metabolism, lipid homeostasis, energy control, and inflammatory responses steps all of critical significance throughout tissue restoration. Accordingly, individually tailored nutritional therapies based on an athlete's genetic signature may enhance muscle repair, prevent inflammation, and optimize training returns (Aruoma et al., 2019; Irani et al., 2025). New confirmation through recent integrative reviews also supports that gene polymorphisms for GSTM1 and SOD2 directly affect an individual's antioxidant potential and directly regulate post-exercise recovery and resistance against oxidative stress. Depending on the genotype, these polymorphisms suggest that targeted nutritional therapies, such as targeted antioxidant supplements, may enable better recovery patterns and potentially reduce the length of time between return-to-play intervals (Guo et al., 2024).

Worth noting is that future potential combines nutrigenomic testing with wearable and artificial intelligence platforms may permit personalised, time-specific changes in both micronutrient and macronutrient consumption throughout training and recovery phases. This approach aligns with the emerging Gene × Diet × Exercise precision model proposed for elite athlete management (Borzemska et al., 2024).

While the potential of this approach is promising, its translation into clinical practice faces several challenges. Most studies are conducted on ethnically homogenous populations, limiting the generalizability of findings. The interpretation of complex genetic data also poses a significant hurdle for sports medicine practitioners. Finally, careful consideration of ethical issues, particularly data privacy, is essential. Overcoming these limitations through multi-ethnic research and robust ethical frameworks will be crucial for the full realization of precision sports medicine (Penggalih et al., 2025; Bojarczuk, 2024). In summary, this section demonstrates that integrating genomic insights into "genotype-driven nutrition" and "training load calibration" can significantly enhance injury prevention and recovery strategies. This paradigm supports the transition from generalized, population-based protocols to a truly "individualized performance medicine" approach in elite athletes (Racinais et al., 2023). The conceptual model (Fig. 1) illustrates the integrative role of genetic and environmental factors in modulating injury risk and recovery efficiency.

8. Ethical, Legal, and practical considerations

As genetic tools become increasingly integrated into sports medicine, ethical, legal, and practical implications must be carefully addressed. The implementation of genomic screening, PRS, and nutrigenomic profiling in athletic settings raises several concerns that extend beyond

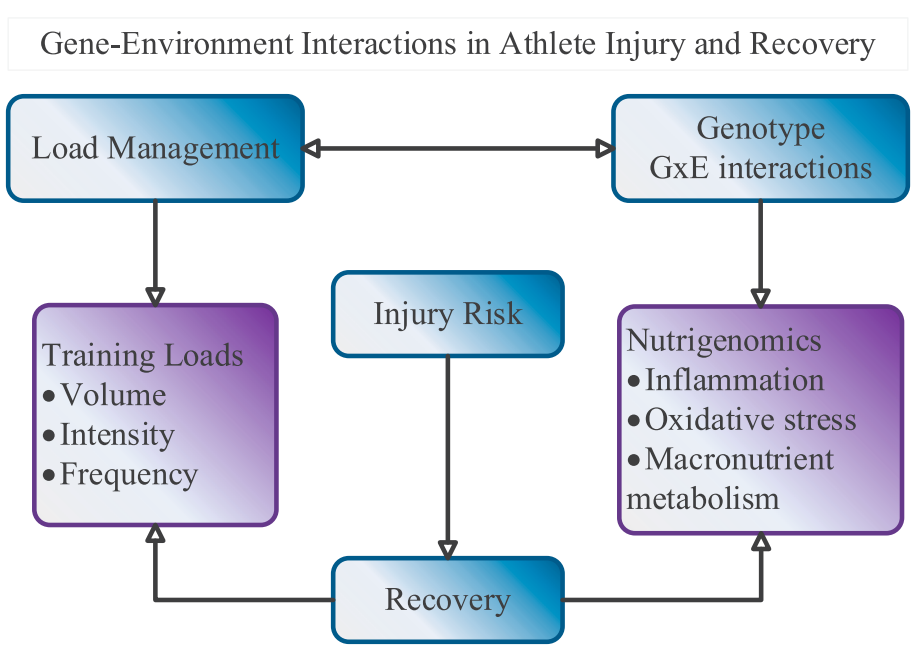

Fig. 1. Schematic representation of gene–environment (G × E) interactions in injury management and recovery in elite athletes.

scientific validity (Williams et al., 2016).

8.1. Privacy and genetic discrimination

One of the foremost concerns is the risk of genetic data misuse. Athletes may feel pressured to undergo genetic testing or disclose their genomic profiles to coaches, sponsors, or sports organizations, potentially leading to genetic discrimination or exclusion based on perceived biological risk or performance potential. Ensuring confidentiality and autonomy in genetic decision-making is essential, as emphasized by emerging sports bioethics literature (Vlahovich et al., 2017). Current ethical frameworks in elite sport recommend robust data protection measures and informed consent protocols, akin to those in clinical genomics. However, regulatory enforcement remains inconsistent across organizations and countries, increasing the vulnerability of athletes to misuse or misinterpretation of genetic data (Bojarczuk, 2024).

8.2. Validity and Equity across ethnic populations

Many genetic associations identified in GWAS and PRS models are based primarily on European-ancestry cohorts, limiting their generalizability to ethnically diverse athletic populations. This lack of representation may result in biased predictions and unjust outcomes in talent selection, training decisions, or injury prevention protocols. The scientific community is increasingly recognizing the need for inclusive, multiethnic genomic databases to improve the fairness and accuracy of genetic profiling in global sports settings. Without equitable representation, there is a risk of widening disparities between athletes from different genetic backgrounds, inadvertently reinforcing structural inequities in access to science-based interventions (Johnson and Pasaniuc, 2024; Addressing, 2022).

8.3. Challenges in coaching and clinical integration

Even when scientifically valid, translating genomic findings into practical decision-making in sport remains challenging (Psatha et al., 2025). Most coaches and clinicians lack the training to interpret genetic results, while genetic counselors are rarely integrated into athlete support teams (Taranto et al., 2018). This knowledge gap increases the risk of misinterpreting PRS or SNP data, potentially leading to inappropriate training loads, misguided return-to-play decisions, or biased talent identification (Williams et al., 2016). Moreover, the predictive value of individual SNPs remains modest, and environmental influences such as training, nutrition, and psychosocial stress frequently outweigh genetic predisposition. Therefore, genomic information should be considered complementary to rather than a replacement for, functional assessments, medical evaluation, and continuous performance monitoring (Moran and Wang, 2020).

8.4. Ethical use of genetic enhancement

The potential extension of genetic technologies from injury prevention to performance enhancement raises profound ethical concerns. Although no current technology allows for direct “gene doping,” the theoretical possibility exists. Emerging gene-editing platforms such as CRISPR-Cas9, as well as epigenetic modulation strategies, could in principle create future avenues for enhancement beyond natural limits. Consequently, both the World Anti-Doping Agency (WADA) and the scientific community consider genetic manipulation a major ethical and regulatory challenge in sports medicine (Naumann et al., 2023; Li et al., 2024; Varillas-Delgado et al., 2022).

8.5. Conclusion of ethical perspective

To ethically integrate genomics into sports medicine, it is essential to balance scientific innovation with the protection of athletes’ rights, dignity, and well-being. Establishing transparent, evidence-based frameworks requires multidisciplinary oversight, uniting geneticists, ethicists, clinicians, and athlete advocates to safeguard against misuse while enabling responsible translation of genomic insights into practice (Williams et al., 2016).

9. Future Directions and clinical translation

As genetic research in sports medicine advances, translating scientific findings into real-world athletic care remains a priority. The future of elite athlete management lies in integrating genomics with rehabilitation sciences, digital health technologies, and precision coaching (Dergaa and Chamari, 2024).

9.1. Personalized rehabilitation protocols

Genotype-informed rehabilitation offers the potential to optimize recovery timelines and reduce the risk of re-injury. For instance, athletes with polymorphisms in collagen-related genes (e.g., COL1A1, COL5A1) may benefit from extended recovery periods and tailored eccentric loading strategies due to slower tissue remodeling. Likewise, muscle fiber type-altering changes in the ACTN3 gene might harbor worthwhile potential for personalizing certain types and intensities of physiotherapeutic methods. Epigenetic biomarkers, such as DNA methylation and miRNA patterns, are capable of reporting in real-time about tissue recovery and adaptation and thus enable person-specific therapeutic intervention (Aali and Shirkani, 2024; Plaza-Diaz et al., 2022; Abdolmaleki et al., 2020). Recent studies lend credence to the argument, with exercise modalities and intensities inducing varying epigenetic responses in muscle and circulation tissue. Such epigenetic modifications, such as DNA methylation and miRNA expression modifications, carry potential as markers for optimizing and personalizing exercise programs in the future. Nonetheless, as noted by the study, there is a need for better understanding of exercise-evoked epigenetic modifications in targeted tissue and their implications and thus for undertaking other studies considering other variables including age throughout the human life span (Zimmer et al., 2016).

9.2. Genetic screening for injury prevention

PRS and SNP screening are expected to constitute key elements in pre-participation screening paradigms. Through an assessment of total genetic risk for musculoskeletal injury, sport clinicians are capable of stratifying athletes into differential risk categories and actively adjusting training volume, stimulus, and exposure (Goodlin et al., 2015). Growing evidence suggests that PRS including variants in COL1A1, MMP3, GDF5, and IL6 genes exhibits superior predictive potential over single-gene-based analysis. However, the extensive application of such screening tools demands validation in multiple-ethnicity cohorts and concordance with sport-specific demands. This recommendation is equally applicable in other fields of research. For instance, an independent study evaluating the inclusion of polygenic scores (PGSs) in the Fracture Risk Assessment Tool (FRAX) for the prediction of hip fractures and major osteoporotic fractures (MOF) in postmenopausal women observed that the scores failed significantly to improve the predictive ability for MOF and provided merely marginal gains in the prediction of hip fractures. These outcomes highlight the fact that merely the integration of genetic data may be insufficient to substantially improve risk predictions in numerous scenarios and instead suggest the necessity for future research in disclosing other determiners (Xiao and Wu, 2023). Such variability highlights the extreme necessity for contextual validation. For instance, in a better application, a frailty-related PRS was validated across two independent populations, the Lothian Birth Cohort 1936 and the English Longitudinal Study of Ageing. The study observed that it significantly predicted frailty at all time points measured, explaining 2.1 % and 1.6 % of the variance in the respective populations. This study exemplifies the

potential of scrupulously validated PRS as informative tools for the identification of individuals susceptible to certain conditions (Flint et al., 2024).

9.3. Wearable technology and artificial intelligence integration

The intersection of genomics and wearable sensors and artificial intelligence (AI) holds the potential for a new age of precision athlete tracking (Srivastava et al., 2024). Wearables may be used to constantly measure biomechanical stress, sleeping, heart rate variability, and recovery biomarkers and supply environmental context about genetically-conferred susceptibilities. AI algorithms may then combine this information with an athlete's genomic profile and identify injury risk trajectories, optimize load management, and recommend individualized recovery protocols (Musat et al., 2024; Spanakis et al., 2025). Recent studies suggest that convolutional and recurrent neural networks are capable of combining heterogeneous data streams, such as those from wearables-derived physiological parameters, biomechanical measurements, and video-based tracking and formulate performance trends, track fatigue, and forecast injury risk with greater precision than conventional monitoring strategies (Jain, 2025). Despite the enormous potential of integrating AI with sport, it is challenged by a number of key problems. First and foremost is an issue of data privacy and wider ethical issues. With AI systems analyzing enormous datasets of sensitive athlete data, responsible data management and transparency are key in order to establish trust. Moreover, there are practical considerations about embedding adoption of these technologies with sport organisations and seamless integration with established praxis need consideration. These considerations are vital in order to ensure that the potential of AI is turned into practical and fair solutions in tangible settings (Zhou et al., 2025). For instance, future AI-based decision-support systems may be capable of combining ACTN3 genotype data with concurrent indicators of muscle fatigue and epigenetic indicators of stress and dynamically customize training prescriptions. In addition, machine learning algorithms trained on large-scale athlete datasets are promising tools for the identification of hitherto unknown gene-environment-outcome interactions and thus may improve predictive precision in optimizing performance and preventing injury (Zhang and Li, 2022; Rebelo et al., 2023).

The future of sport genomics is multidimensionally integrated and integrates static genetic data with dynamic physiological, behavioral, and contextual information. With interdisciplinary work, ethical guidelines, and technologically driven application, genomics will become increasingly informative about personal, adaptive, and evidence-based care for elite athletes (Barh, 2020).

10. Conclusion

The field of sport genomics is developing rapidly, and it has brought with it unrivaled knowledge of biological substrates linked with susceptibility to injuries and variations in recovery in elite athletes. In this review, it is established that gene variations, PRSs, and epigenetic mechanisms collectively act on tissue robustness, inflammation resolution, and oxidative stress tolerance (Clos et al., 2019; Yazdan Nasab et al., 2025).

PRS have emerged as more robust predictors than single-gene approaches, especially when integrated with wearable biosensors and AI-based platforms for dynamic athlete monitoring. These advances highlight the promise of moving from generalized, population-based injury-prevention models to anticipatory, personalized interventions (Voisin et al., 2015; da Silva et al., 2020). Nonetheless, several challenges must be addressed for clinical translation. These include the limited validity of current genetic findings across diverse ethnic populations, the modest predictive power of individual variants, and the interpretative gap between genomic insights and real-world coaching or clinical decision-making (Popejoy and Fullerton, 2016; Pickering et al., 2019). Looking ahead, the advancement of this field requires the development of multi-ethnic genomic resources, the conduct of large-scale longitudinal studies in athlete populations, and the integration of these data with digital health technologies and behavioral analytics. Simultaneously, adherence to ethical and legal principles, including data privacy protection and safeguarding athlete autonomy, is essential for the responsible implementation of these approaches (Gusliana et al., 2024).

In conclusion, the transition toward precision sports medicine will require a multidisciplinary approach uniting genomics, bioinformatics, physiology, and behavioral sciences. By embracing the complexity of gene-environment interactions and adopting a systems biology perspective, the field can progress toward evidence-based, personalized strategies that enhance performance, extend career longevity, and safeguard athlete well-being at the highest level of competition.

Author contributions

Hao Ding participated in data collection and analysis. Qilu Deng contributed to methodology development and manuscript drafting. Zhenhua Guo, as the corresponding author, supervised the project, reviewed the results, and finalized the manuscript. All authors approved the final version of the paper.

Ethical approval

The study does not involve experiments with human participants or animals, and therefore ethical approval was not applicable.

CRediT authorship contribution statement

Hao Ding: Writing – review & editing, Writing – original draft, Resources, Investigation, Data curation. **Qilu Deng:** Writing – review & editing, Writing – original draft, Visualization, Funding acquisition, Formal analysis. **Zhenhua Guo:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Conceptualization.

Funding

This research did not receive any external funding.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors have no acknowledgments to declare.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.gene.2025.149957.

Data availability

Data will be made available on request.

Data related to this study can be obtained from the corresponding author upon reasonable request.

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